

Task 2: Motor Speed Control Using PWM and Obstacle Detection

1. Project Overview

Motor speed control and obstacle detection are two fundamental concepts widely used in robotics, industrial automation, autonomous vehicles, and smart transportation systems. This project demonstrates how an Arduino Uno can control the speed of a DC motor using Pulse Width Modulation (PWM) while simultaneously detecting obstacles using an ultrasonic sensor. The Arduino continuously measures the distance between the sensor and nearby objects. If an obstacle is detected within a predefined distance, the motor speed is reduced or stopped to prevent collision. This project provides practical knowledge of motor control, sensor interfacing, and real-time decision-making using embedded systems.

2. Aim / Objective

The objective of this project is to design and implement a motor control system that regulates the speed of a DC motor using PWM signals and detects nearby obstacles using an ultrasonic sensor. Based on the measured distance, the Arduino automatically adjusts the motor speed to ensure safe operation and avoid collisions.

3. Components Required

- Arduino Uno
- DC Motor
- L293D Motor Driver IC
- HC-SR04 Ultrasonic Sensor
- Breadboard
- Jumper Wires
- External Power Supply (if required)

4. Software Required

- Arduino IDE
- Tinkercad Circuits Simulator

5. Theory

Pulse Width Modulation (PWM) is a widely used technique for controlling the speed of DC motors by varying the average voltage supplied to the motor. Instead of continuously changing the voltage level, PWM rapidly switches the motor's power ON and OFF at a fixed frequency. By changing the percentage of time the signal remains ON, known as the duty cycle, the average power delivered to the motor changes, thereby controlling its speed. A higher duty cycle results in higher motor speed, while a lower duty cycle reduces the speed.

The HC-SR04 ultrasonic sensor is used to measure the distance between the system and nearby obstacles. It operates by transmitting ultrasonic sound waves and measuring the time taken for the reflected waves to return after striking an object. The Arduino calculates the distance using the speed of sound and continuously monitors the surroundings.

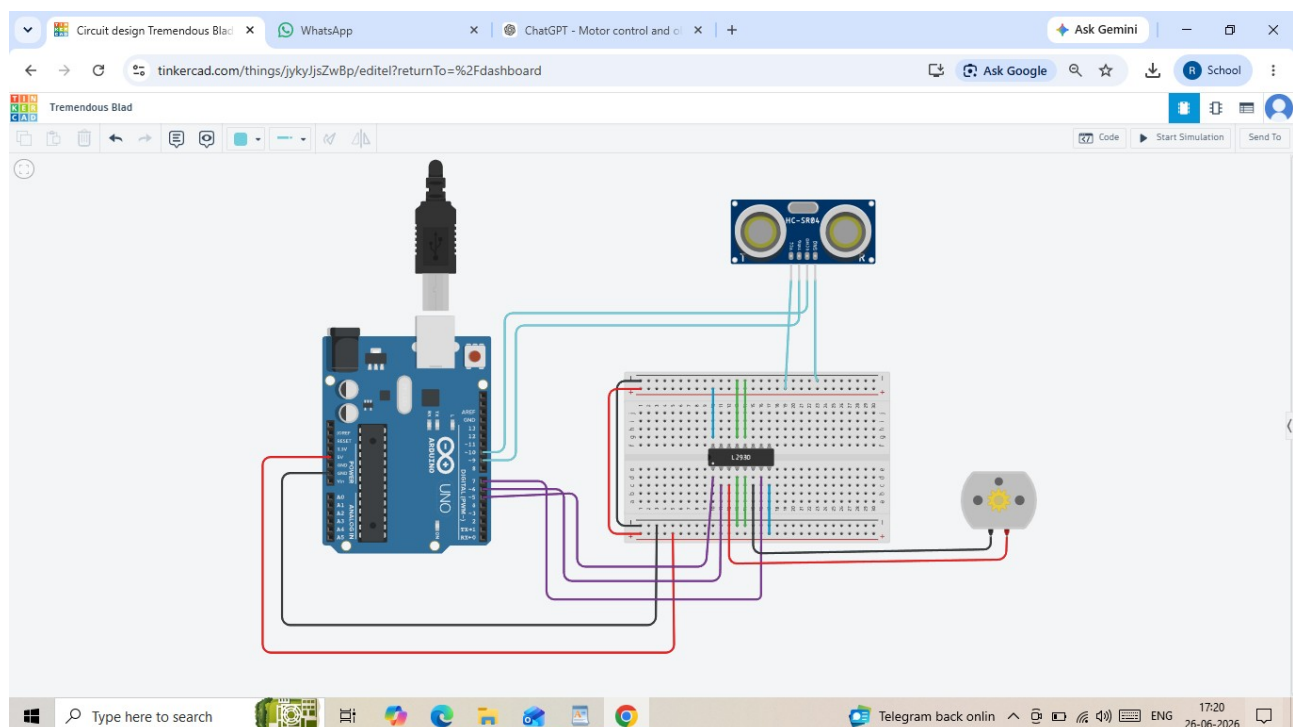
Since the Arduino cannot directly drive a DC motor due to its limited current output, an L293D motor driver is used as an interface between the Arduino and the motor. The motor driver receives PWM signals from the Arduino and supplies sufficient current to operate the motor safely.

Whenever the ultrasonic sensor detects an obstacle within a specified distance, the Arduino processes the sensor data and reduces the PWM duty cycle or completely stops the motor. This automatic control mechanism improves operational safety and demonstrates the integration of sensing and control in embedded systems.

6. Working Principle

The Arduino continuously triggers the ultrasonic sensor to measure the distance to nearby objects. The sensor sends ultrasonic pulses and receives the reflected echoes, allowing the Arduino to calculate the distance. Based on the measured distance, the Arduino generates a PWM signal to control the motor speed through the L293D motor driver. If no obstacle is detected, the motor runs at the predefined speed. When an obstacle is detected within the safety range, the Arduino automatically reduces the motor speed or stops the motor to prevent collision. This process repeats continuously, enabling real-time monitoring and motor control.

Connections:



Source code :

```
const int trigPin = 9;
const int echoPin = 10;

const int enA = 5;
const int in1 = 6;
const int in2 = 7;

long duration;
int distance;

void setup()
{
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  pinMode(enA, OUTPUT);
  pinMode(in1, OUTPUT);
  pinMode(in2, OUTPUT);

  Serial.begin(9600);

  // Motor Forward
  digitalWrite(in1, HIGH);
  digitalWrite(in2, LOW);
}

void loop()
{
  // Send ultrasonic pulse
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);

  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);

  distance = duration * 0.034 / 2;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  if(distance < 20)
  {
    analogWrite(enA, 0);

    Serial.println("Obstacle Detected");
```

```
    Serial.println("Motor Stopped");
}
else
{
    analogWrite(enA, 200);

    Serial.println("Motor Running");
}

delay(500);
}
```

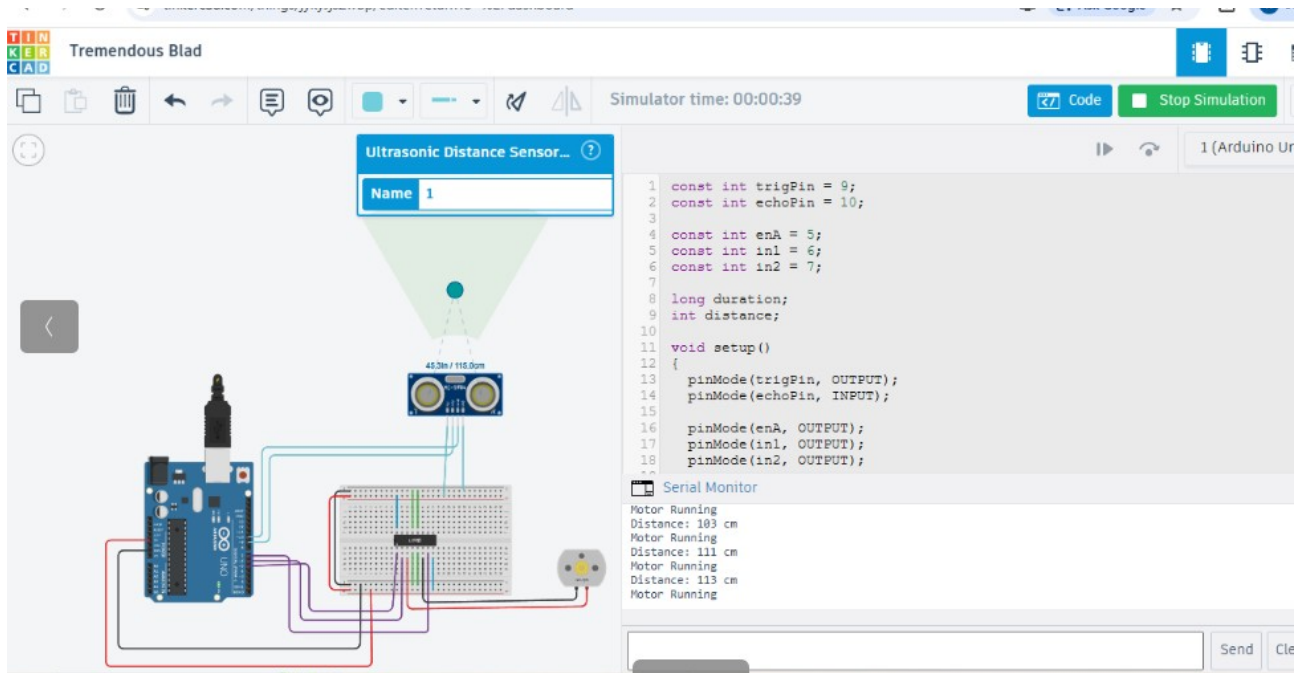
7. Algorithm

1. Start the program.
2. Initialize the ultrasonic sensor and motor driver.
3. Configure the PWM output pin for motor speed control.
4. Trigger the ultrasonic sensor to measure the distance.
5. Read the echo pulse and calculate the distance.
6. Compare the measured distance with the predefined threshold.
7. If the obstacle is farther than the threshold, run the motor at normal speed.
8. If the obstacle is within the threshold distance, reduce the motor speed or stop the motor.
9. Repeat the process continuously.

8. How to Run

1. Open the project in Tinkercad.
2. Verify all component connections.
3. Upload the Arduino program to the virtual Arduino Uno.
4. Click **Start Simulation**.
5. Adjust the position of the obstacle in front of the ultrasonic sensor.
6. Observe how the motor speed changes according to the measured distance.
7. Monitor the distance values through the Serial Monitor if included in the program.

Output in serial monitor:



9. Applications

- Autonomous robots
- Obstacle avoidance systems
- Automatic guided vehicles (AGVs)
- Smart parking systems
- Industrial conveyor belt control
- Robotics projects
- Automated transportation systems
- Educational embedded system projects

10. Advantages

- Provides automatic obstacle detection.
- Enables smooth motor speed control using PWM.
- Improves operational safety by preventing collisions.
- Simple and cost-effective implementation.
- Easily expandable for advanced robotic applications.
- Suitable for both educational and industrial automation projects.

11. Skills Gained

- Arduino programming using C/C++.
- Pulse Width Modulation (PWM) programming.
- Ultrasonic sensor interfacing.

- DC motor control using L293D motor driver.
- Distance measurement techniques.
- Real-time decision-making in embedded systems.
- Circuit simulation using Tinkercad.

12. Future Scope

The project can be enhanced by integrating additional sensors such as infrared sensors, cameras, or LiDAR for more accurate obstacle detection. Wireless communication technologies such as Wi-Fi or Bluetooth can be added to enable remote monitoring and control. Machine learning algorithms can also be integrated to improve obstacle recognition and autonomous navigation. The system can further be expanded into fully autonomous robotic vehicles, smart wheelchairs, warehouse automation, and industrial mobile robots.

13. Conclusion

The Motor Speed Control Using PWM and Obstacle Detection project successfully demonstrates the integration of motor control techniques with real-time sensing using an Arduino Uno. The system automatically adjusts the motor speed based on the distance measured by the ultrasonic sensor, improving safety and operational efficiency. This project provides practical experience in PWM control, sensor interfacing, motor driver operation, and embedded system programming. It serves as an excellent foundation for developing advanced robotic and automation systems.

Internship :

Completed as part of Embedded Systems Internship – Intern spark.